WORKED SOLUTION

STEP II 1988, Question 16

Probability and Statistics • Normal Distribution

STEP II 1988, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q16

The key idea

Everything about the roots is controlled by the single random variable B, so each question about X_1 and X_2 should be turned into a question about which interval B lies in. The discriminant tells us when the roots are real, and Vieta's formulas (sum $-2B$, product 1) make the conditions in (i) and (ii) easy to translate. After that, each answer is a conditional probability or conditional expectation for a standard normal variable, and the expectation needs one short integral.

Solution

Throughout, $B \sim \mathrm{N}(0,1)$, with density $\phi(z) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-z^2/2}$ and distribution function Φ. Let R be the event that the roots are real.

Real roots

The discriminant of $X^2 + 2BX + 1$ is $4B^2 - 4$, so the roots are real exactly when $B^2 \geq 1$, that is, $|B| \geq 1$. By symmetry of the normal distribution,

$$\mathrm{P}(R) = \mathrm{P}(|B| \geq 1) = 2(1 - \Phi(1)) = 2(1 - 0.84134) = 0.31731\ldots \approx 0.317.$$

(Whether we include the equal-root case $|B| = 1$ makes no difference, since it has probability zero.)

When the roots are real they are

$$X_{1,2} = -B \pm \sqrt{B^2 - 1}, \qquad X_1 + X_2 = -2B, \qquad X_1 X_2 = 1.$$

Part (i)

Ask yourself How do we turn "both roots exceed $\frac{1}{5}$" into a condition on B?

First, both roots must be positive. Their product is $1 > 0$, so they have the same sign, and that sign is the sign of their sum $-2B$. So we need $B < 0$, and with R this means $B \leq -1$. Write $b = -B \geq 1$; the roots are then $b \pm \sqrt{b^2 - 1}$, both positive.

Since the product is 1, the smaller root is the reciprocal of the larger one:

$$b - \sqrt{b^2 - 1} = \frac{1}{b + \sqrt{b^2 - 1}}.$$

So the smaller root exceeds $\frac{1}{5}$ if and only if the larger root is less than 5, and then the larger root automatically exceeds $\frac{1}{5}$ as well. The condition is therefore

$$\sqrt{b^2 - 1} < 5 - b.$$

This needs $b < 5$, and for $1 \leq b < 5$ both sides are non-negative, so we may square without changing the inequality:

$$b^2 - 1 < 25 - 10b + b^2 \iff 10b < 26 \iff b < 2.6.$$

(Values $b \geq 5$ fail anyway, as $2.6 < 5$.) Hence, given real roots, both roots exceed $\frac{1}{5}$ exactly when $-2.6 < B \leq -1$, and

$$\begin{aligned} \mathrm{P}(X_1, X_2 > \tfrac{1}{5} \mid R) &= \frac{\mathrm{P}(-2.6 < B \leq -1)}{\mathrm{P}(|B| \geq 1)} = \frac{\Phi(2.6) - \Phi(1)}{2(1 - \Phi(1))} \\ &= \frac{0.99534 - 0.84134}{0.31731} = 0.4853\ldots \approx 0.485. \end{aligned}$$

Part (ii)

Since $|X_1 + X_2| = 2|B|$, we want $\mathrm{E}(2|B| \,|\, |B| \geq 1)$. For a conditional expectation given an event, we integrate over the event and divide by its probability:

$$\mathrm{E}(|B| \,|\, |B| \geq 1) = \frac{1}{\mathrm{P}(|B| \geq 1)} \int_{|z| \geq 1} |z|\, \phi(z)\, \mathrm{d}z = \frac{2\int_1^\infty z\, \phi(z)\, \mathrm{d}z}{2(1 - \Phi(1))},$$

using the symmetry of ϕ. The integral is easy because $z\,\phi(z)$ is minus the derivative of $\phi(z)$:

$$\int_1^\infty z\, \phi(z)\, \mathrm{d}z = \Big[-\phi(z)\Big]_1^\infty = \phi(1) = \frac{\mathrm{e}^{-1/2}}{\sqrt{2\pi}} = 0.24197\ldots$$

Therefore

$$\mathrm{E}(|X_1 + X_2| \,|\, R) = \frac{2\phi(1)}{1 - \Phi(1)} = \frac{2 \times 0.24197}{0.15866} = 3.050\ldots \approx 3.05.$$

Checks

- Try $B = -2.6$: the equation is $X^2 - 5.2X + 1 = 0$, with roots 5 and $\frac{1}{5}$ exactly, the boundary case in (i).
- The answer to (ii) must exceed 2, because $2|B| \geq 2$ on R. It does, and it is plausible: the normal tail beyond 1 is thin, so $|B|$ is usually only a little above 1 (its conditional mean is about 1.53).
- A simulation of 2×10^6 standard normal values of B gave 0.3177, 0.4855 and 3.049, agreeing with 0.317, 0.485 and 3.05.

Common slips

- Forgetting to condition: in (i) and (ii) we must divide by $\mathrm{P}(R) = 0.317$. The unconditional probability $\Phi(2.6) - \Phi(1) = 0.154$ is not the answer.
- Including $B \geq 1$ in (i). Both roots are then negative, so neither exceeds $\frac{1}{5}$.
- Computing $\mathrm{E}|X_1 + X_2|$ as $|\mathrm{E}(X_1 + X_2)|$, which is 0 by symmetry.

Beyond STEP

The quantity $\phi(1)/(1 - \Phi(1))$ is the mean of a *truncated* normal distribution, and the ratio $\phi(a)/(1 - \Phi(a))$ is known as the inverse Mills ratio; it appears in statistics (for example, in models of censored data) and in reliability theory as the hazard rate of the normal distribution. Questions about random polynomials, such as how many real roots a polynomial with random coefficients has on average, form a surprisingly rich area of probability. For conditional expectation and continuous distributions, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

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